International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Mid Exam, Spring 2024

Course Code: CSE 3631 **Course Title**: Operating System

Time: 1 hours 30 minutes **Full Marks**: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1.	a)	Describe the interrupt handling process in an operating system.	**C01**	03
	b)	How does a user program make a system call?	**CO1**	03
	c)	Explain the benefits of multiprogramming and multitasking. How do they improve system efficiency and user experience? Or List five services provided by an operating system, and explain how each creates convenience for users.	**C01**	04
2.	a)	Imagine a busy internet cafe with 5 customers: **Alice** (Priority 3): Needs to check her email quickly (short burst). **Bob** (Priority 2): Working on a presentation (medium burst). **Charlie** (Priority 1): Urgent online banking transaction (short burst). **David** (Priority 3): Playing an online game (long burst). **Eve** (Priority 2): Researching for a project (medium burst). The scheduling algorithm : Preemptive Priority-based, round-robin scheduling algorithm Process queue: Initially, all processes are in the queue (Alice, Bob, Charlie, David, Eve). CPU Time Slice: 2 seconds Short Burst :4 seconds Medium Burst : 7 seconds Long Burst : 10 seconds Now draw the Gantt chart , calculate the average waiting time and CPU utilization.	**CO2**	07
	b)	Compare and contrast different approaches to operating system structures: Microkernel, Layered and Hybrid.	**C02**	03
3.	a)	Compare the performance implications of context switching in different time quantum. **Or**	**CO2**	02

Investigate the impact of multithreading on performance and resource utilization.

b) Imagine a busy computer lab with four students waiting to use the printer. Schedule the students using Shortest Remaining Time First (SRTF) Scheduling Algorithm. Calculate the average waiting time. **CO2** 04

Student:	**Arrival Time:**	**Burst Time (Initial):**	**Burst Time (Remaining):**
Alice	10:00 AM	20 minutes	20 minutes
Bob	10:02 AM	10 minutes	10 minutes
Charlie	10:05 AM	30 minutes	30 minutes
David	10:10 AM	5 minutes	5 minutes

c) Imagine a self-driving car with three critical tasks: **C02** 04

Sensor Data Acquisition (**Task A**): Reads data from various sensors (e.g., camera, LiDAR) every 10 milliseconds (ms). Deadline: 10 ms.
Obstacle Detection (**Task B**): Analyzes sensor data for potential hazards every 50 ms. Deadline: 50 ms.
Path Planning (**Task C**): Calculates the optimal driving path based on obstacles every 100 ms. Deadline: 100 ms.

Illustrate the scheduling of theses process using Rate Monotonic Scheduling algorithm.
Or
Schedule the above scenario using Earliest Deadline First (EDF) algorithm.

//// The End/////